

IoT Home Automation Digital Logic

Sensor/actuator core with UART register interface, clock-gated sleep controller and an ESP8266 bridge verified through a Python mock

AT A GLANCE

40 / 40 checks PASS

System test

16 / 16 replies match

Mock bridge exchange

-23.6 % (7.06 -> 5.40 uW)

Dynamic power (sleep)

-53.8 %

Net toggles (sleep)

-16.1 % (9.45 -> 7.93 uW)

Total power

968 LC (12 %), 43.3 MHz

iCE40 HX8K (nextpnr)

674 LUT4, 414 FF

Yosys synth_ice40

464 LUT, 427 FF

Xilinx xc7 (stat)

VERDICT

SIM PASS

POWER -16 %

SYNTH OK

TIMING MET

UNITTEST 4/4

NO HW RUN

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Objective and brief mapping

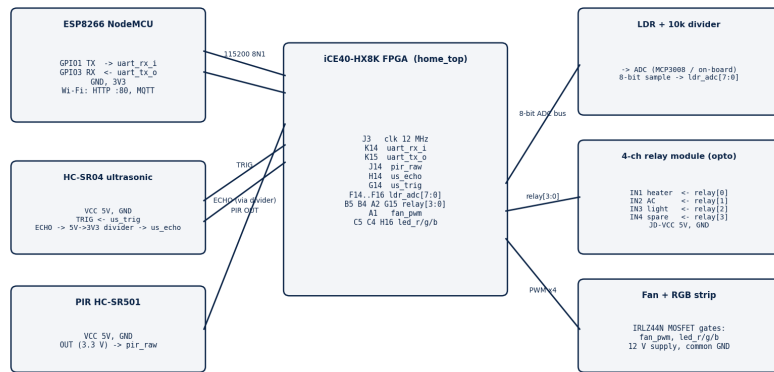
The brief: a Verilog home-automation core with PIR debounce/one-shot, LDR hysteresis comparator on an 8-bit ADC value, HC-SR04 echo timer, relays with interlock, 8-bit fan PWM and RGB PWM, a register-mapped ASCII UART command interface (SET R1 1 / GET T style), clock gating of idle blocks with a sleep controller (lower PWM/ADC activity after N seconds without motion) and wake-on-PIR, a testbench covering sensors, commands, relays, PWM duty and sleep entry/exit, an activity-based power comparison always-on vs sleep-enabled with tools/power_estimate.py, synth_ice40 + nextpnr stats, an ESP8266 HTTP+MQTT bridge sketch, a Python mock bridge talking to the simulation through a file/pipeline UART model with a unit-tested shared parser, a wiring diagram, and the declaration that no ESP8266/HC-05 hardware was run. All items were implemented; the sketch is provided but not flashed.

Architecture

Block	Function	Domain
uart_rx / uart_tx / cmd_regs	8N1 UART, line parser SET/GET, register file, reply formatter (OK / ERR / =)	always on
pir_oneshot	2-flop sync, 1 ms debounce, motion level + one-clock pulse: the wake source	always on
sleep controller	1 s tick, idle-seconds counter vs register N; asleep when idle >= N; cleared by motion or SET W 1	always on
clock_gate -> pwm8 x4, ultrasonic_timer, LDR sample timer	negedge-registered enable AND clk: glitch-free gated clock that stops during sleep	gated (gclk)
ldr_hyst	dark below 80, light above 120; sampled every 10 ms awake, once per second asleep	always on, rate switched
relay_ctrl	R0 heater / R1 AC interlock (later request refused, I counter), R2 follows dark when A=1	always on

IoT home-automation core -- wiring diagram (3.3 V logic on the FPGA side, 5 V sensors level-shifted)

Declared: none of this hardware was used; the diagram documents the intended build for the ESP8266 bridge + FPGA core.



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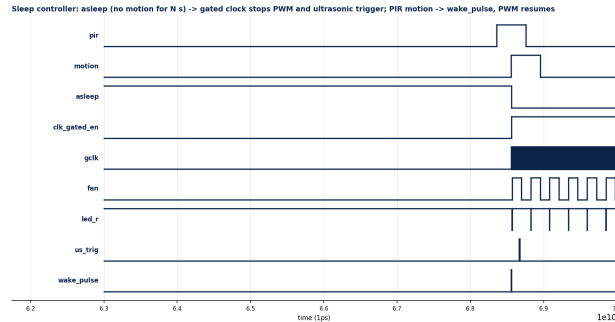
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Area	What was checked	Result
Sleep / wake	SET N 5; motion; idle -> asleep, clk_gated_en 0, trigger stopped, fan frozen; GET Z=001; PIR -> awake < 0.4 ms, fan 50 % again; asleep again; SET W 1 wakes	PASS
Mock bridge	16 HTTP/MQTT-derived commands via file UART model; 16/16 replies equal the Python register model (incl. 3 ERR cases)	PASS
Totals	40/40 checks SLEEP_EN=1, 31/31 SLEEP_EN=0, 7/7 in each gate-level power run, unittest 4/4	PASS



Sleep entry and wake: asleep high, gated clock and PWM/trigger stopped; PIR motion -> wake_pulse, clk_gated_en, gclk and fan PWM resume.

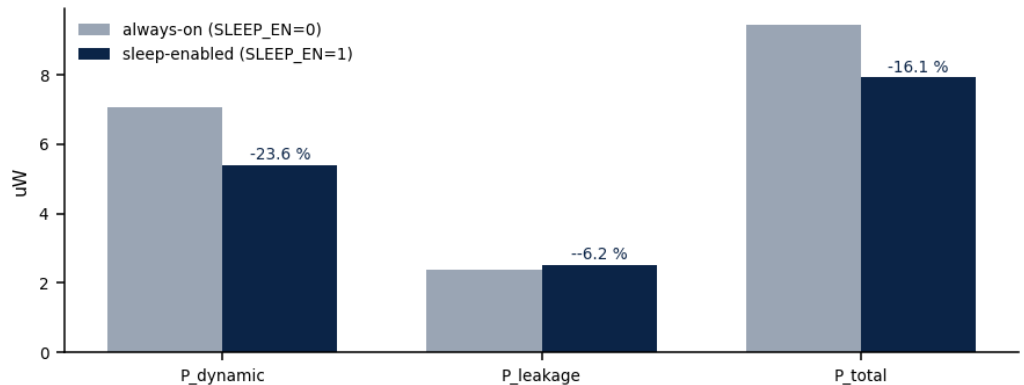
Power: always-on vs sleep-enabled (gate level, identical stimulus, 50.3 ms, ~80 % idle)

The clock net is the largest consumer in both variants; gating removes it only from the ~120 flops of the gated domain, so dynamic power falls less than toggles. My first scheme (PWM gated, timers on enables, mostly-awake stimulus) was 7 % worse than always-on; moving the ultrasonic and LDR timers into the gated domain and measuring on an idle-heavy scenario gave the saving above.

Most active nets confirm the mechanism: in both runs clk toggles 100 618 times (50.3 ms at 1 MHz) and the always-on netlist has six counter LSBs toggling every cycle (50 307 each: PWM and timer counters). In the sleep-enabled run those bits are gone; the busiest nets after clk are the clock-gate inverter (100 618), sec_cnt[0] (48 806) and gclk itself with 27 628 toggles, i.e. the gated clock ran for only 27 % of the simulation. By driving cell type the DFF-driven (Q output) power falls from 0.522 to 0.217 uW and NOR/NAND logic from 0.59 to 0.25 uW, while the AND-driven share rises (0.21 -> 0.53 uW) because gclk is now an AND output feeding 130 clock pins.

Metric	Always-on	Sleep-enabled	Change
Gate netlist (cmos_cells.lib, sim config)	2485 cells, 17 352 um2	2675 cells, 18 385 um2	+7.6 % area
Net toggles	3 239 651	1 495 562	-53.8 %
P_dynamic	7.061 uW	5.396 uW	-23.6 %
P_leakage	2.385 uW	2.532 uW	+6.2 %
P_total	9.447 uW	7.928 uW	-16.1 %

Gate-level activity power, same stimulus (50.3 ms, ~80 % idle)



Synthesis and implementation (real 12 MHz configuration)

Flow (run here)	Measured result
Yosys 0.68 synth_ice40	1445 cells: 674 SB_LUT4, 414 FF (269 DFFESR, 117 DFFSR, 14 DFFESS, 11 DFF, 2 DFFSS, 1 DFFN), 345 SB_CARRY
nextpnr-ice40 hx8k ct256, --freq 12 --seed 1	968 / 7680 LC (12 %), 54 SB_IO, 3 SB_GB; Fmax 43.32 MHz on clk (PASS at 12 MHz), 147.8 MHz on gclk; critical path in cmd_regs reply formatter
icetime -d hx8k	total path delay 23.83 ns (41.96 MHz)
Yosys synth_xilinx xc7 (stat)	1051 cells: 464 LUT2-6, 427 FF (410 FDRE, 1 FDRE_1, 16 FDSE), 75 CARRY4, 6 MUXF78

KEY TAKEAWAY Gating the PWM, ultrasonic and LDR-timer clocks when no motion is seen cut gate-level toggles by 54 % and dynamic power by 24 % on an idle-heavy stimulus; the UART register interface was driven end to end by a Python Wi-Fi mock.

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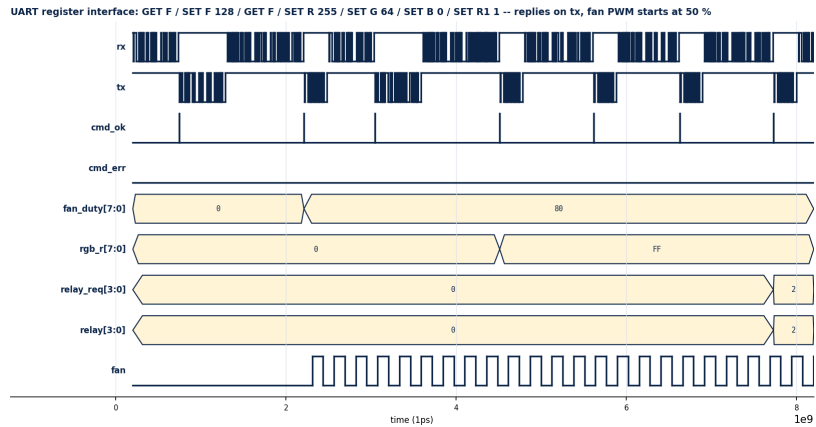
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UART command traffic and wake-on-PIR detail

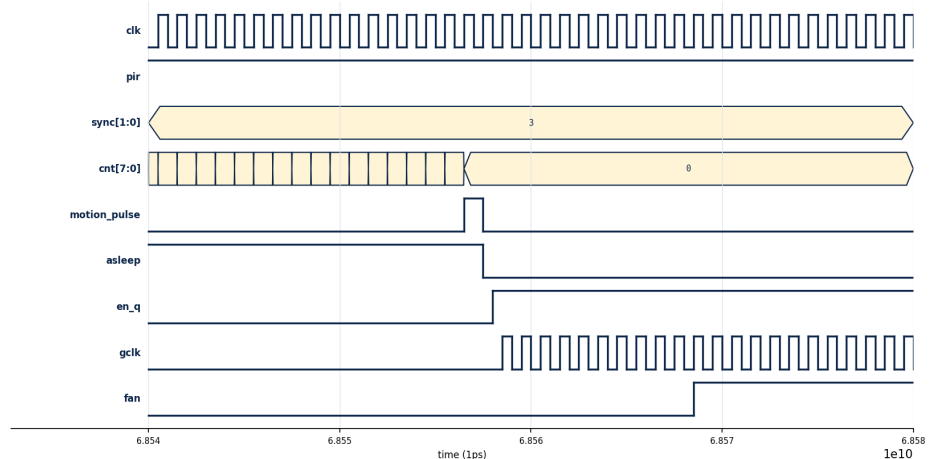


First seven commands on rx with their replies on tx; fan_duty becomes 128 and the fan PWM starts at 50 %, relay_req/relay follow SET R1 1 (cmd_ok pulses per accepted line).

Per-module resources (synth_xilinx hierarchy) and wake-up path

Module	FF	LUT	CARRY4	Note
cmd_regs	168	276	22	parser, 8-byte reply buffer, /100 and /10 formatters: the critical path
ultrasonic_timer	77	70	15	20-bit period counter (60 ms at 12 MHz), 16-bit width and distance
uart_rx + uart_tx	54	50	6	104-clock bit timers
home_top glue	58	27	16	sleep tick counter (24 bit), idle seconds, LDR sample timer
pir_oneshot	18	7	4	1 ms debounce counter
4 x pwm8 + relay_ctrl + ldr_hyst + clock_gate	52	34	12	gated domain = 4 x 9 FF + ultrasonic 77 + LDR timer 17 = about 130 of 427 FF

Wake-on-PIR zoom: 2-flop sync, debounce count, motion_pulse clears asleep, en_q re-armed on the falling clock edge, gclk restarts glitch-free



Wake-on-PIR zoom (40 us): raw pir through the 2-flop sync, debounce count to 200, motion_pulse clears asleep, en_q re-arms on the falling edge and gclk restarts without a partial pulse.

Declared substitutions / not run here

- No ESP8266/HC-05, PIR, HC-SR04, LDR/ADC, relay board or FPGA was used. esp8266_bridge.ino (ESP8266WiFi, ESP8266WebServer, PubSubClient) was not compiled or flashed; its grammar and request mapping are the ones exercised by cmd_protocol.py against the RTL.
- The mock-bridge 'pipe' is a batch file exchange (uart_cmds.txt -> RTL UART -> uart_resp.txt), not a live socket: Icarus here has no VPI socket bridge.
- Power values use the educational liberty and power_estimate.py on the sim-scaled configuration: relative comparison only. Xilinx numbers are Yosys estimates (no Vivado); the iCE40 bitstream was not programmed; Icarus Verilog instead of ModelSim.

Learnings / next steps

Power optimisation must be measured: the first gating scheme increased power until whole timer domains were gated and an idle-heavy stimulus was used. dist is a SystemVerilog keyword (Icarus rejected the testbench until renamed). Next: a DHT22 bit-banged decoder, a live socket UART model (cocotb/VPI), a gated domain for the UART transmitter, and flashing the ESP8266 sketch to close the loop.

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